PM Group Work Project 1 M3 Submission / GWP_1_Portfolio_Management.ipynb - Colab.pdf

WorldQuant University

Document Details

Submission ID

trn:oid:::8628:554317369

Submission Date

Feb 8, 2026, 6:21 PM GMT+3

Download Date

Feb 8, 2026, 6:30 PM GMT+3

File Name

GWP_1_Portfolio_Management.ipynb - Colab.pdf

File Size

959.3 KB

33 Pages

6,636 Words

40,205 Characters

13% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Small Matches (less than 30 words)

Exclusions

- 3 Excluded Sources

Match Groups

16 Not Cited or Quoted 13%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

4% Internet sources
2% Publications
13% Submitted works (Student Papers)

Match Groups

- **16** Not Cited or Quoted 13%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

4% Internet sources
2% Publications
13% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	Houston Community College on 2025-06-28	2%
2	Submitted works	WorldQuant University on 2026-01-13	2%
3	Submitted works	WorldQuant University on 2025-10-08	1%
4	Submitted works	WorldQuant University on 2025-11-04	<1%
5	Submitted works	University of Lancaster on 2025-04-02	<1%
6	Internet	medium.com	<1%
7	Internet	github.com	<1%
8	Submitted works	SKEMA Business School on 2025-10-03	<1%
9	Submitted works	University of California, Los Angeles on 2023-10-19	<1%
10	Submitted works	WorldQuant University on 2025-10-23	<1%

11 Submitted works

Temple University on 2021-04-13 <1%

12 Submitted works

Universidad TecMilenio on 2025-11-30 <1%

13 Submitted works

WorldQuant University on 2025-10-23 <1%

14 Submitted works

WorldQuant University on 2026-01-13 <1%

15 Submitted works

Johns Hopkins Unversity on 2025-04-22 <1%

PART 1

```
import yfinance as yf
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import minimize


tickers = ['TSLA', 'WMT', 'BAC', 'GS', 'LLY', 'MRK', 'GOOG', 'META', 'AAPL', 'XOM']
start_date = '2023-07-01'
end_date = '2025-06-30'
oos_start = '2025-07-01'
oos_end = '2025-09-30'


df = yf.download(tickers, start=start_date, end=oos_end, auto_adjust=True)


if 'Close' in df.columns:
    data = df['Close']
else:
    data = df

train_data = data.loc[start_date:end_date]
oos_data = data.loc[oos_start:oos_end]


returns = train_data.pct_change().dropna()
mean_returns = returns.mean() * 252
cov_matrix = returns.cov() * 252


def portfolio_performance(weights, mean_returns, cov_matrix):
    p_ret = np.dot(weights, mean_returns)
```

turnitin Page 6 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
    p_std = np.sqrt(np.dot(weights.T, np.dot(cov_matrix, weights)))
    return p_std, p_ret

def neg_sharpe_ratio(weights, mean_returns, cov_matrix, risk_free_rate=0.04):
    p_std, p_ret = portfolio_performance(weights, mean_returns, cov_matrix)
    return -(p_ret - risk_free_rate) / p_std

def solve_optimal_weights(mean_returns, cov_matrix, max_weight=1.0):
    num_assets = len(mean_returns)
    args = (mean_returns, cov_matrix)
    constraints = ({'type': 'eq', 'fun': lambda x: np.sum(x) - 1})
    bounds = tuple((0, max_weight) for _ in range(num_assets))


    init_guess = num_assets * [1./num_assets]

    result = minimize(neg_sharpe_ratio, init_guess,
                      args=args, method='SLSQP', bounds=bounds, constraints=constraints)
    return result.x


weights_step1 = solve_optimal_weights(mean_returns, cov_matrix, max_weight=1.0)


weights_step2 = solve_optimal_weights(mean_returns, cov_matrix, max_weight=0.15)


def get_frontier(mean_returns, cov_matrix, max_weight):
    # Create a range of target returns between the minimum and maximum mean returns
    target_returns = np.linspace(mean_returns.min(), mean_returns.max(), 30)
    frontier_volatility = []

    for target in target_returns:
        cons = ({'type': 'eq', 'fun': lambda x: np.sum(x) - 1},
                {'type': 'eq', 'fun': lambda x: np.dot(x, mean_returns) - target})
        bounds = tuple((0, max_weight) for _ in range(len(mean_returns)))
        res = minimize(lambda x: np.sqrt(np.dot(x.T, np.dot(cov_matrix, x))),
                       len(mean_returns) * [1./len(mean_returns)],
                       method='SLSQP', bounds=bounds, constraints=cons)
```

turnitin Page 6 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 7 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

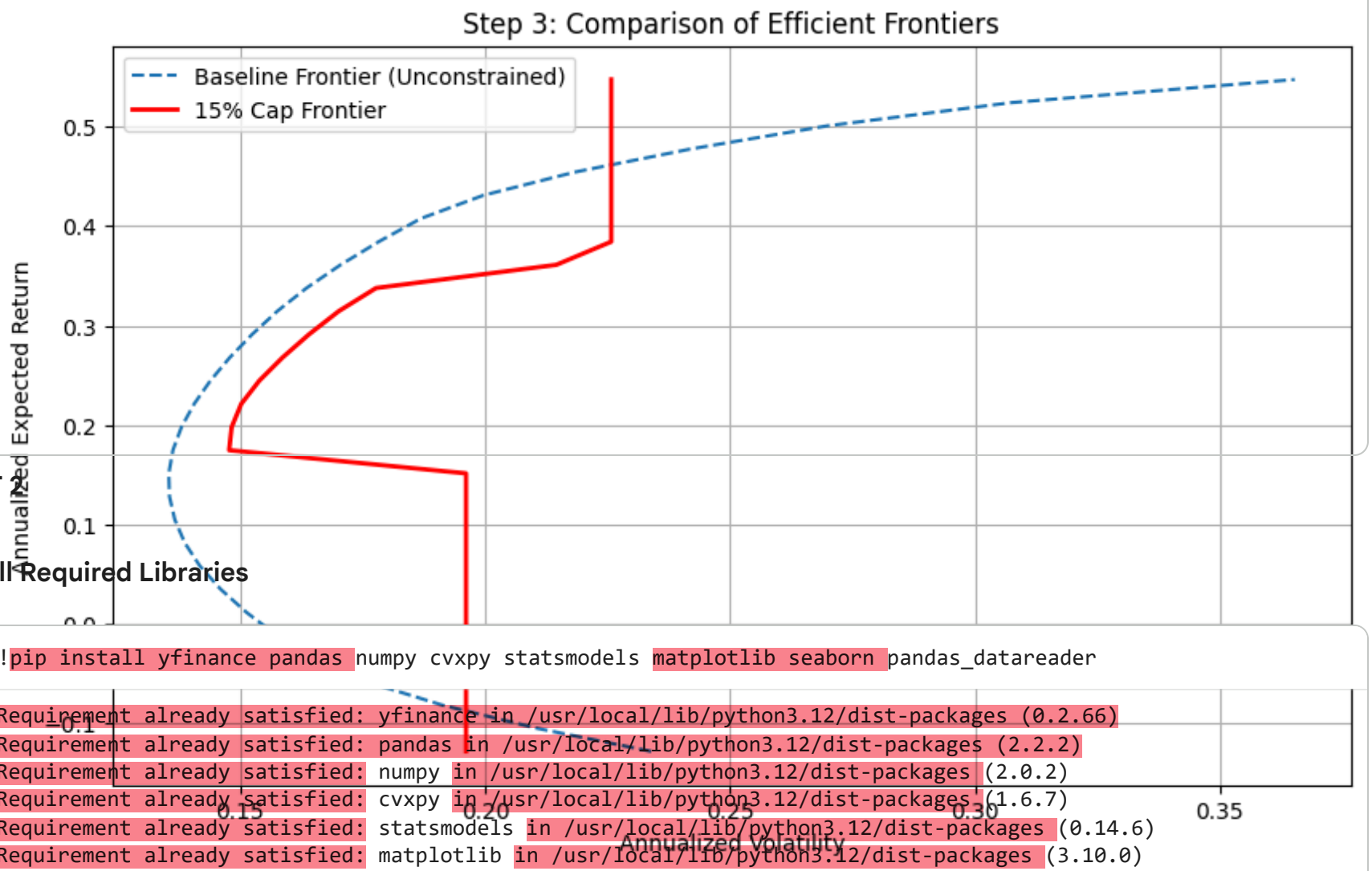
```
    frontier_volatility.append(res.fun)
    return frontier_volatility, target_returns

vol1, ret1 = get_frontier(mean_returns, cov_matrix, 1.0)
vol2, ret2 = get_frontier(mean_returns, cov_matrix, 0.15)

plt.figure(figsize=(10,6))
plt.plot(vol1, ret1, label='Baseline Frontier (Unconstrained)', linestyle='--')
plt.plot(vol2, ret2, label='15% Cap Frontier', color='red', linewidth=2)
plt.title('Step 3: Comparison of Efficient Frontiers')
plt.xlabel('Annualized Volatility')
plt.ylabel('Annualized Expected Return')
plt.grid(True)
plt.legend()
plt.show()

oos_returns = oos_data.pct_change().dropna()
port1_daily_rets = (oos_returns * weights_step1).sum(axis=1)
port2_daily_rets = (oos_returns * weights_step2).sum(axis=1)

print("-" * 30)
print("Step 1 Weights (Unconstrained):")
for t, w in zip(tickers, weights_step1):
    print(f"{t}: {w:.2%}")

print("\nStep 2 Weights (15% Cap):")
for t, w in zip(tickers, weights_step2):
    print(f"{t}: {w:.2%}")

print("-" * 30)
print(f"Step 1 OOS Cumulative Return: {((1 + port1_daily_rets).prod() - 1):.2%}")
print(f"Step 2 OOS Cumulative Return: {((1 + port2_daily_rets).prod() - 1):.2%}")
```

turnitin Page 7 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 8 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369
turnitin Page 8 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 9 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
[**********************100%***********************]  10 of 10 completed
```



PART 2

Install Required Libraries

```
!pip install yfinance pandas numpy cvxpy statsmodels matplotlib seaborn pandas_datareader
```

```
Requirement already satisfied: yfinance in /usr/local/lib/python3.12/dist-packages (0.2.66)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: cvxpy in /usr/local/lib/python3.12/dist-packages (1.6.7)
Requirement already satisfied: statsmodels in /usr/local/lib/python3.12/dist-packages (0.14.6)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: pandas_datareader in /usr/local/lib/python3.12/dist-packages (0.10.0)
Requirement already satisfied: requests>=2.31 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2.32.
Requirement already satisfied: multitasking>=0.0.7 in /usr/local/lib/python3.12/dist-packages (from yfinance) (
Requirement already satisfied: platformdirs>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from yfinance) (
Requirement already satisfied: pytz>=2022.5 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2025.2)
Requirement already satisfied: frozendict>=2.3.4 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2.
Requirement already satisfied: peewee>=3.16.2 in /usr/local/lib/python3.12/dist-packages (from yfinance) (3.19.
Requirement already satisfied: beautifulsoup4>=4.11.1 in /usr/local/lib/python3.12/dist-packages (from yfinance
```

turnitin Page 9 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 10 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
Requirement already satisfied: curl_cffi>=0.7 in /usr/local/lib/python3.12/dist-packages (from yfinance) (0.14.
Requirement already satisfied: protobuf>=3.19.0 in /usr/local/lib/python3.12/dist-packages (from yfinance) (5.2
Requirement already satisfied: websockets>=13.0 in /usr/local/lib/python3.12/dist-packages (from yfinance) (15.
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: osqp>=0.6.2 in /usr/local/lib/python3.12/dist-packages (from cvxpy) (1.1.0)
Requirement already satisfied: clarabel>=0.5.0 in /usr/local/lib/python3.12/dist-packages (from cvxpy) (0.11.1)
Requirement already satisfied: scs>=3.2.4.post1 in /usr/local/lib/python3.12/dist-packages (from cvxpy) (3.2.11
Requirement already satisfied: scipy>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from cvxpy) (1.16.3)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels) (1.0.
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from statsmodels) (2
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3
Requirement already satisfied: lxml in /usr/local/lib/python3.12/dist-packages (from pandas_datareader) (6.0.2)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4>=4
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beauti
Requirement already satisfied: cffi in /usr/local/lib/python3.12/dist-packages (from clarabel>=0.5.0->cvxpy) (2
Requirement already satisfied: certifi>=2024.2.2 in /usr/local/lib/python3.12/dist-packages (from curl_cffi>=0.
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from osqp>=0.6.2->cvxpy) (3.1
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from osqp>=0.6.2->cvxpy)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from osqp>=0.6.2->cvxpy) (1.5
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from reques
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.31->yf
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi->clarabel>=0.5.0
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->osqp>=0
```

1. Import Libraries

```
import numpy as np
import pandas as pd
import yfinance as yf
import cvxpy as cp
import statsmodels.api as sm
```

turnitin Page 10 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 11 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
import matplotlib.pyplot as plt
import seaborn as sns
from pandas_datareader import data as web
```

2. Download Stock Price Data (July 2023 – June 2025)

```
tickers = ['TSLA','WMT','BAC','GS','LLY','MRK','GOOG','META','AAPL','XOM']

start_date = "2023-07-01"
end_date = "2025-06-30"

prices = yf.download(
    tickers,
    start=start_date,
    end=end_date,
    auto_adjust=True
)

returns = prices.pct_change().dropna()
returns.head()
```

15

turnitin Page 11 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 12 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
[*********************100%***********************]  10 of 10 completed
```

Price	Close										..
Ticker	AAPL	BAC	GOOG	GS	LLY	META	MRK	TSLA	WMT	XOM	..
Date											
2023-07-05	-0.005871	-0.004110	0.017170	-0.020085	0.009276	0.029194	-0.005510	0.009506	-0.000569	-0.005118	.
2023-07-06	0.002509	-0.027510	-0.013863	-0.022028	-0.005347	-0.008085	-0.017151	-0.021028	-0.006325	-0.037321	.
2023-07-07	-0.005891	0.008840	-0.006533	0.006933	-0.024028	-0.005000	-0.024608	-0.007630	-0.023041	0.002332	.
2023-07-10	-0.010856	0.004557	-0.027218	0.004125	0.002389	0.012288	0.008807	-0.017564	0.009056	0.014831	.
2023-07-11	-0.002810	0.012561	0.007187	0.013935	-0.030387	0.014247	-0.011459	0.000668	-0.001485	0.012227	.

5 rows × 50 columns

3. Compute Expected Returns & Covariance Matrix

```
mu = returns.mean()
Sigma = returns.cov()

mu, Sigma
```

```
(Price   Ticker
 Close   AAPL      0.000257
         BAC       0.001206
         GOOG      0.000974
         GS        0.001767
         LLY       0.001310
         META      0.002164
         MRK      -0.000510
```

turnitin Page 12 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 13 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
         TSLA    0.001081
         WMT     0.001365
         XOM     0.000275
High     AAPL    0.000227
         BAC     0.001203
         GOOG    0.000947
         GS      0.001732
         LLY     0.001341
         META    0.002128
         MRK    -0.000530
         TSLA    0.000879
         WMT     0.001353
         XOM     0.000252
Low      AAPL    0.000242
         BAC     0.001230
         GOOG    0.000915
         GS      0.001755
         LLY     0.001330
         META    0.002129
         MRK    -0.000529
         TSLA    0.000934
         WMT     0.001356
         XOM     0.000247
Open     AAPL    0.000278
         BAC     0.001260
         GOOG    0.000951
         GS      0.001801
         LLY     0.001360
         META    0.002204
         MRK    -0.000510
         TSLA    0.001148
         WMT     0.001366
         XOM     0.000302
Volume   AAPL    0.058109
         BAC     0.054664
         GOOG    0.072575
         GS      0.070105
         LLY     0.094397
         META    0.072834
         MRK     0.090217
         TSLA    0.030576
         WMT     0.074048
```

turnitin Page 13 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 14 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
          XOM      0.051001
dtype: float64,
Price             Close                                                  \
Ticker             AAPL       BAC          GOOG        GS       LLY
Price  Ticker
Close  AAPL    0.000301  0.000098  1.485862e-04  0.000130  0.000083
       BAC     0.000098  0.000273  7.802768e-05  0.000233  0.000044
```

4. Mean–Variance Optimization (Max Weight = 15%)

```
close_returns = returns["Close"]
```

```
mu_close = close_returns.mean()
Sigma_close = close_returns.cov()
```

```
import cvxpy as cp
import pandas as pd

n = len(mu_close)
w = cp.Variable(n)

objective = cp.Minimize(cp.quad_form(w, Sigma_close))

constraints = [
    cp.sum(w) == 1,
    w >= 0,
    w <= 0.15
]

problem = cp.Problem(objective, constraints)
problem.solve()

optimal_weights = pd.Series(w.value, index=mu_close.index)

optimal_weights.sort_values(ascending=False)
```

turnitin Page 14 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 15 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

Ticker	0
BAC	1.500000e-01
GOOG	1.500000e-01
WMT	1.500000e-01
XOM	1.500000e-01
MRK	1.500000e-01
AAPL	1.193168e-01
LLY	1.186119e-01
GS	1.123714e-02
META	8.341653e-04
TSLA	-4.495229e-20

dtype: float64

5. Compute Portfolio Returns

```
close_returns = returns["Close"]
```

```
close_returns = close_returns[optimal_weights.index]
```

```
portfolio_returns = close_returns.dot(optimal_weights)
portfolio_returns.head()

print(close_returns.columns.tolist())
print(optimal_weights.index.tolist())
```

turnitin Page 15 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 16 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
['AAPL', 'BAC', 'GOOG', 'GS', 'LLY', 'META', 'MRK', 'TSLA', 'WMT', 'XOM']
['AAPL', 'BAC', 'GOOG', 'GS', 'LLY', 'META', 'MRK', 'TSLA', 'WMT', 'XOM']
```

6. Download Fama–French 5-Factor Data

```
ff_factors = web.DataReader('F-F_Research_Data_5_Factors_2x3_daily',
                            'famafrench', start=start_date)[0]

ff_factors.index = pd.to_datetime(ff_factors.index)

ff_factors = ff_factors / 100
ff_factors.head()
```

```
/tmp/ipython-input-825446906.py:1: FutureWarning: The argument 'date_parser' is deprecated and will be removed
  ff_factors = web.DataReader('F-F_Research_Data_5_Factors_2x3_daily',
```

Date	Mkt-RF	SMB	HML	RMW	CMA	RF
2023-07-03	0.0015	0.0027	0.0106	0.0018	0.0011	0.0002
2023-07-05	-0.0029	-0.0106	-0.0026	-0.0030	-0.0001	0.0002
2023-07-06	-0.0094	-0.0058	-0.0069	0.0057	0.0005	0.0002
2023-07-07	-0.0013	0.0130	0.0121	-0.0023	0.0018	0.0002
2023-07-10	0.0036	0.0124	-0.0001	-0.0103	0.0054	0.0002

7. Align Portfolio & Factor Data

turnitin Page 16 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 17 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
data = pd.concat([portfolio_returns, ff_factors], axis=1).dropna()

data.columns = ['Portfolio'] + list(ff_factors.columns)

data.head()
```

Date	Portfolio	Mkt-RF	SMB	HML	RMW	CMA	RF
2023-07-05	0.000478	-0.0029	-0.0106	-0.0026	-0.0030	-0.0001	0.0002
2023-07-06	-0.015915	-0.0094	-0.0058	-0.0069	0.0057	0.0005	0.0002
2023-07-07	-0.009931	-0.0013	0.0130	0.0121	-0.0023	0.0018	0.0002
2023-07-10	0.000550	0.0036	0.0124	-0.0001	-0.0103	0.0054	0.0002
2023-07-11	-0.000916	0.0076	0.0018	0.0048	-0.0021	0.0028	0.0002

8. Compute Excess Portfolio Returns

```
data['Excess_Portfolio'] = data['Portfolio'] - data['RF']
```

9. OLS Regression (Fama–French 5-Factor Model)

```
X = data[['Mkt-RF','SMB','HML','RMW','CMA']]
X = sm.add_constant(X)
Y = data['Excess_Portfolio']

ols_model = sm.OLS(Y, X).fit()
print(ols_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:       Excess_Portfolio   R-squared:                       0.708
Model:                            OLS   Adj. R-squared:                  0.705
```

turnitin Page 17 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 18 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
Method:                 Least Squares   F-statistic:                     238.5
Date:                Tue, 03 Feb 2026   Prob (F-statistic):          5.71e-129
Time:                        16:40:44   Log-Likelihood:                 1929.1
No. Observations:                 498   AIC:                            -3846.
Df Residuals:                     492   BIC:                            -3821.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       -1.88e-05      0.000     -0.082      0.934      -0.000       0.000
Mkt-RF         0.8019      0.024     32.886      0.000       0.754       0.850
SMB           -0.1056      0.039     -2.734      0.006      -0.181      -0.030
HML            0.2047      0.039      5.256      0.000       0.128       0.281
RMW            0.1074      0.053      2.035      0.042       0.004       0.211
CMA           -0.0373      0.053     -0.703      0.483      -0.142       0.067
==============================================================================
Omnibus:                       27.671   Durbin-Watson:                   1.967
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               72.014
Skew:                           0.217   Prob(JB):                     2.30e-16
Kurtosis:                       4.812   Cond. No.                         271.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

10. Robust Regression (HC3 Standard Errors)

```
robust_model = sm.OLS(Y, X).fit(cov_type='HC3')
print(robust_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:       Excess_Portfolio   R-squared:                       0.708
Model:                            OLS   Adj. R-squared:                  0.705
Method:                 Least Squares   F-statistic:                     125.5
Date:                Tue, 03 Feb 2026   Prob (F-statistic):           1.80e-85
Time:                        16:40:48   Log-Likelihood:                 1929.1
No. Observations:                 498   AIC:                            -3846.
Df Residuals:                     492   BIC:                            -3821.
```

turnitin Page 18 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 10 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
Df Model:                           5
Covariance Type:                  HC3
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       -1.88e-05      0.000     -0.081      0.935      -0.000       0.000
Mkt-RF         0.8019      0.035     22.956      0.000       0.733       0.870
SMB           -0.1056      0.048     -2.204      0.028      -0.200      -0.012
HML            0.2047      0.056      3.648      0.000       0.095       0.315
RMW            0.1074      0.066      1.615      0.106      -0.023       0.238
CMA           -0.0373      0.078     -0.480      0.631      -0.190       0.115
==============================================================================
Omnibus:                       27.671   Durbin-Watson:                   1.967
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               72.014
Skew:                           0.217   Prob(JB):                     2.30e-16
Kurtosis:                       4.812   Cond. No.                         271.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

11. Training / Testing Split (Time-Series Safe)

```
split_date = "2025-01-01"

train = data.loc[:split_date]
test = data.loc[split_date:]

X_train = sm.add_constant(train[['Mkt-RF','SMB','HML','RMW','CMA']])
Y_train = train['Excess_Portfolio']

X_test = sm.add_constant(test[['Mkt-RF','SMB','HML','RMW','CMA']])
Y_test = test['Excess_Portfolio']

train_model = sm.OLS(Y_train, X_train).fit()

predictions = train_model.predict(X_test)
```

turnitin Page 19 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

12. Out-of-Sample Performance Metrics

```
rmse = np.sqrt(((Y_test - predictions)**2).mean())
r2_out = 1 - ((Y_test - predictions)**2).sum() / ((Y_test - Y_test.mean())**2).sum()

rmse, r2_out
```

```
(np.float64(0.006758878938928171), np.float64(0.7767238003005621))
```

13. Correlation Matrix of Fama–French Factors

5

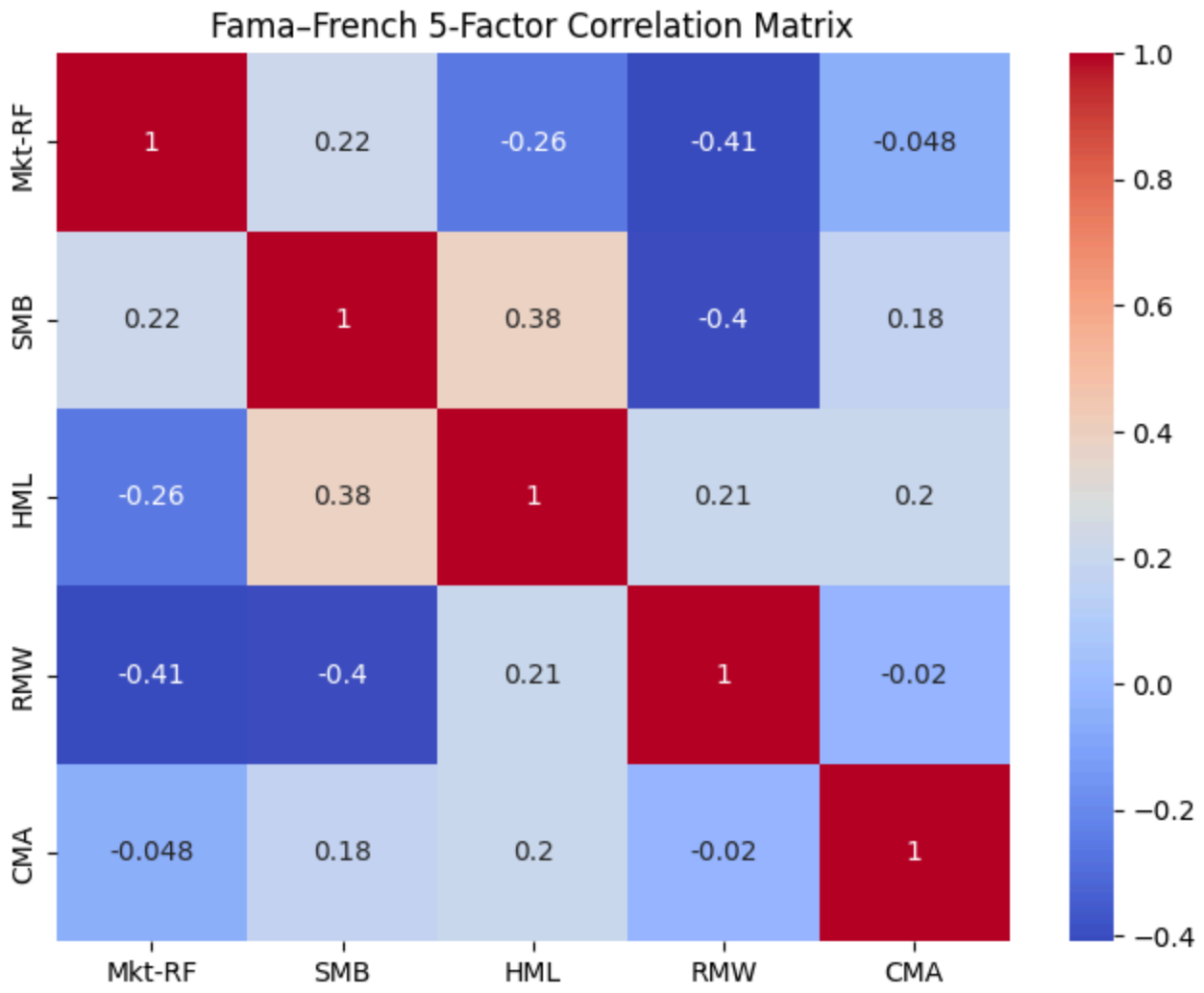
```
corr_matrix = ff_factors[['Mkt-RF','SMB','HML','RMW','CMA']].corr()
corr_matrix
```

	Mkt-RF	SMB	HML	RMW	CMA
Mkt-RF	1.000000	0.216287	-0.257876	-0.408714	-0.048203
SMB	0.216287	1.000000	0.378243	-0.400338	0.175107
HML	-0.257876	0.378243	1.000000	0.209535	0.202981
RMW	-0.408714	-0.400338	0.209535	1.000000	-0.019523
CMA	-0.048203	0.175107	0.202981	-0.019523	1.000000

14. Heatmap Visualization

```
plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Fama-French 5-Factor Correlation Matrix")
plt.show()
```

turnitin Page 21 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

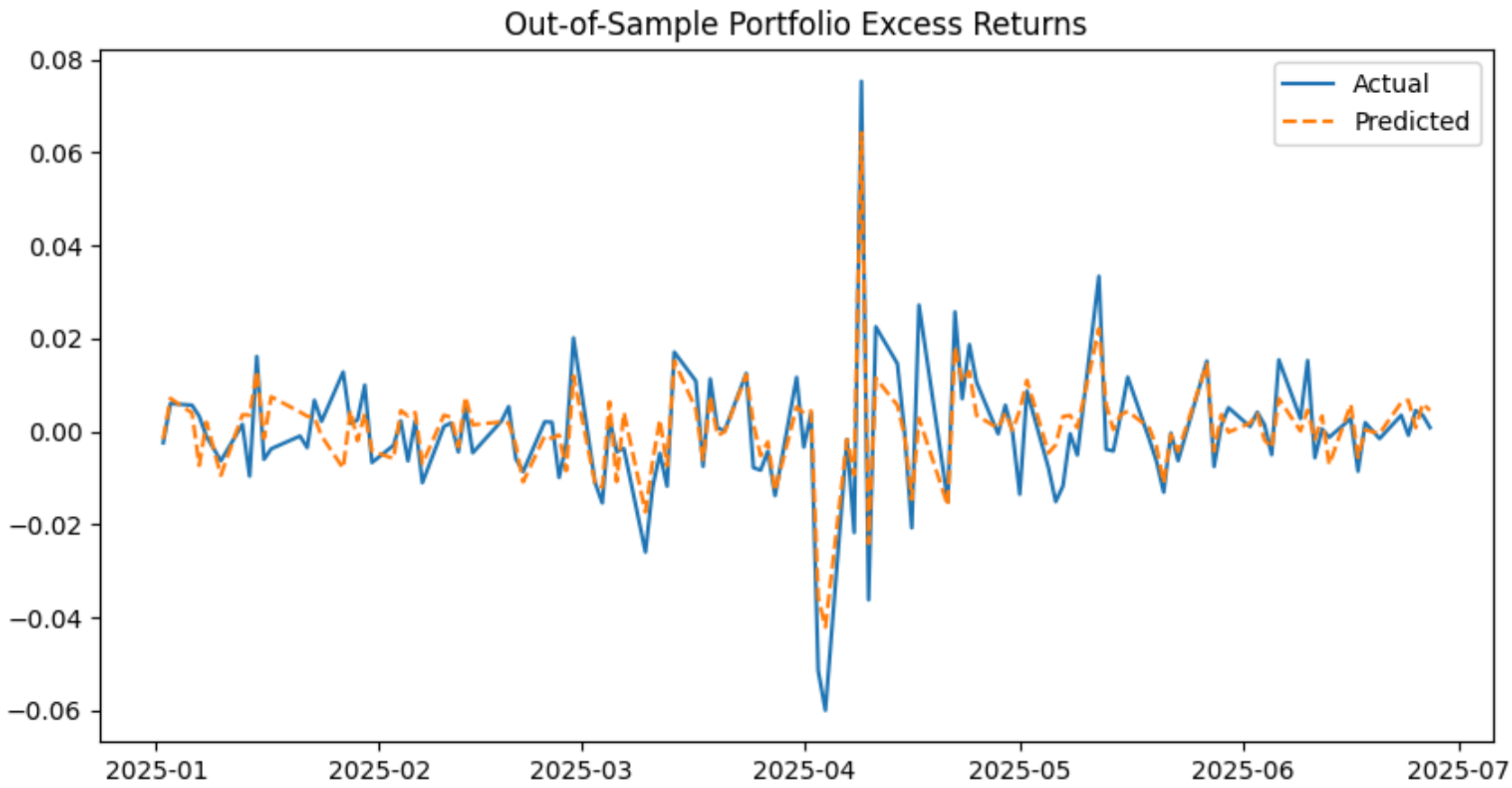


15. Plot Actual vs Predicted Returns

turnitin Page 21 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 22 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
plt.figure(figsize=(10,5))
plt.plot(Y_test.index, Y_test, label="Actual")
plt.plot(predictions.index, predictions, label="Predicted", linestyle="--")
plt.title("Out-of-Sample Portfolio Excess Returns")
plt.legend()
plt.show()
```



16. Export Tables to CSV

```
optimal_weights.to_csv("optimal_weights.csv")
corr_matrix.to_csv("factor_correlation_matrix.csv")
```

turnitin Page 22 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

PART 3.

```
import numpy as np
import pandas as pd
from datetime import datetime
from scipy.optimize import minimize
import warnings
warnings.filterwarnings('ignore')
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (14, 8)

asset_universe = ['TSLA', 'WMT', 'BAC', 'LLY', 'AAPL']
num_assets = len(asset_universe)

period_training_start = datetime(2023, 7, 1)
period_training_end = datetime(2025, 6, 30)

trading_dates_range = pd.bdate_range(start=period_training_start, end=period_training_end)
trading_days_count = len(trading_dates_range)

asset_annual_returns = np.array([0.28, 0.08, 0.12, 0.35, 0.18])
asset_annual_volatilities = np.array([0.65, 0.18, 0.25, 0.48, 0.28])

correlation_matrix = np.array([
    [1.00, 0.15, 0.22, 0.35, 0.40],
    [0.15, 1.00, 0.35, 0.08, 0.25],
    [0.22, 0.35, 1.00, 0.18, 0.32],
    [0.35, 0.08, 0.18, 1.00, 0.28],
    [0.40, 0.25, 0.32, 0.28, 1.00]
])

covariance_matrix_daily = np.outer(asset_annual_volatilities, asset_annual_volatilities) * correlation_matrix

np.random.seed(42)
```

turnitin Page 34 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
daily_log_returns = np.random.multivariate_normal(
    mean=asset_annual_returns / 252,
    cov=covariance_matrix_daily,
    size=trading_days_count
)

initial_price_vector = np.array([200.0, 80.0, 30.0, 600.0, 180.0])
cumulative_return_factors = np.exp(np.cumsum(daily_log_returns, axis=0))
synthetic_price_matrix = initial_price_vector * cumulative_return_factors

price_dataframe = pd.DataFrame(synthetic_price_matrix, columns=asset_universe, index=trading_dates_range)
daily_returns = price_dataframe.pct_change().dropna()

annual_expected_returns = daily_returns.mean() * 252
covariance_matrix_annual = daily_returns.cov() * 252

print("="*80)
print("STEP 7: UNCONSTRAINED OPTIMIZATION VS SIMULATION")
print("="*80)
print("\nAsset Universe:", asset_universe)
print("\nTraining Period: July 1, 2023 - June 30, 2025")
print(f"\nData Points Available: {len(daily_returns)} trading days")
print("\nAnnualized Expected Returns:")
for asset_idx, ticker in enumerate(asset_universe):
    print(f"  {ticker}: {annual_expected_returns.iloc[asset_idx]:.6f} ({annual_expected_returns.iloc[asset_idx
print("\nAnnualized Covariance Matrix:")
print(covariance_matrix_annual)

def portfolio_performance_metrics(weight_allocation, expected_return_vector, cov_matrix_annual):
    portfolio_annual_return = np.dot(weight_allocation, expected_return_vector.values)
    portfolio_annual_volatility = np.sqrt(np.dot(weight_allocation, np.dot(cov_matrix_annual.values, weight_al
    portfolio_sharpe_ratio = portfolio_annual_return / portfolio_annual_volatility if portfolio_annual_volatil
    return portfolio_annual_return, portfolio_annual_volatility, portfolio_sharpe_ratio

def negative_sharpe_objective(weight_allocation, expected_return_vector, cov_matrix_annual):
    portfolio_return, portfolio_volatility, sharpe_metric = portfolio_performance_metrics(weight_allocation, e
    return -sharpe_metric if portfolio_volatility > 0 else 0

constraints_step7 = {'type': 'eq', 'fun': lambda weights: np.sum(weights) - 1.0}
```

turnitin Page 34 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 25 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
bounds_step7 = tuple((0.0, 1.0) for _ in range(num_assets))
initial_weight_guess = np.array([1.0/num_assets] * num_assets)

optimization_solution_step7 = minimize(
    negative_sharpe_objective,
    initial_weight_guess,
    args=(annual_expected_returns, covariance_matrix_annual),
    method='SLSQP',
    bounds=bounds_step7,
    constraints=constraints_step7,
    options={'ftol': 1e-12, 'maxiter': 1000}
)

optimal_weights_step7 = optimization_solution_step7.x
optimal_return_step7, optimal_volatility_step7, optimal_sharpe_step7 = portfolio_performance_metrics(
    optimal_weights_step7, annual_expected_returns, covariance_matrix_annual
)

print("\n" + "="*80)
print("STEP 7: OPTIMIZATION APPROACH")
print("="*80)
print("\nOptimal Portfolio Weights (Optimization):")
for ticker_idx, ticker_name in enumerate(asset_universe):
    print(f"  {ticker_name}: {optimal_weights_step7[ticker_idx]:.6f} ({optimal_weights_step7[ticker_idx]*100:.
print(f"\nOptimal Portfolio Annual Return: {optimal_return_step7:.6f} ({optimal_return_step7*100:.2f}%)")
print(f"Optimal Portfolio Annual Volatility: {optimal_volatility_step7:.6f} ({optimal_volatility_step7*100:.2f
print(f"Optimal Sharpe Ratio: {optimal_sharpe_step7:.6f}")

simulation_sample_sizes_step7 = [10000, 50000, 100000, 250000, 500000, 1000000]
convergence_results_step7 = []

np.random.seed(42)

for sample_size in simulation_sample_sizes_step7:
    random_weights_matrix = np.random.dirichlet(np.ones(num_assets), sample_size)
    simulated_sharpe_values = []

    for random_portfolio_weights in random_weights_matrix:
        sim_port_return, sim_port_volatility, sim_port_sharpe = portfolio_performance_metrics(
```

turnitin Page 25 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 26 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
        random_portfolio_weights, annual_expected_returns, covariance_matrix_annual
        )
        simulated_sharpe_values.append(sim_port_sharpe)

    best_sim_index = np.argmax(simulated_sharpe_values)
    best_simulated_weights = random_weights_matrix[best_sim_index]
    best_sim_return, best_sim_volatility, best_sim_sharpe = portfolio_performance_metrics(
        best_simulated_weights, annual_expected_returns, covariance_matrix_annual
    )

    weight_l1_distance = np.sum(np.abs(best_simulated_weights - optimal_weights_step7))
    sharpe_absolute_difference = np.abs(best_sim_sharpe - optimal_sharpe_step7)
    sharpe_percentage_deviation = (sharpe_absolute_difference / optimal_sharpe_step7) * 100 if optimal_sharpe_

    convergence_results_step7.append({
        'Simulations': sample_size,
        'Optimal_Weights': best_simulated_weights,
        'Portfolio_Return': best_sim_return,
        'Portfolio_Volatility': best_sim_volatility,
        'Sharpe_Ratio': best_sim_sharpe,
        'L1_Distance': weight_l1_distance,
        'Sharpe_Difference': sharpe_absolute_difference,
        'Sharpe_Error_Pct': sharpe_percentage_deviation
    })

    print(f"\nSimulation Count: {sample_size}")
    print(f"  Best Simulated Weights: {[f'{w:.6f}' for w in best_simulated_weights]}")
    print(f"  Best Simulated Sharpe Ratio: {best_sim_sharpe:.6f}")
    print(f"  L1 Distance from Optimal: {weight_l1_distance:.8f}")
    print(f"  Sharpe Ratio Difference: {sharpe_absolute_difference:.8f} ({sharpe_percentage_deviation:.4f}%)")

convergence_table_step7 = pd.DataFrame([
    {
        'Sample_Size': r['Simulations'],
        'L1_Weight_Distance': r['L1_Distance'],
        'Sharpe_Error_Percentage': r['Sharpe_Error_Pct']
    }
    for r in convergence_results_step7
])
```

turnitin Page 26 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
print("\n" + "-"*80)
print("STEP 7: CONVERGENCE ANALYSIS")
print("-"*80)
print(convergence_table_step7.to_string(index=False))

figure_step7, subplot_axes_step7 = plt.subplots(1, 2, figsize=(14, 5))

simulation_counts = [r['Simulations'] for r in convergence_results_step7]
weight_distances = [r['L1_Distance'] for r in convergence_results_step7]
sharpe_errors_percentage = [r['Sharpe_Error_Pct'] for r in convergence_results_step7]

subplot_axes_step7[0].plot(simulation_counts, weight_distances, marker='o', linewidth=2.5, markersize=9, color
subplot_axes_step7[0].set_xscale('log')
subplot_axes_step7[0].set_yscale('log')
subplot_axes_step7[0].set_xlabel('Number of Monte Carlo Simulations', fontsize=12, fontweight='bold')
subplot_axes_step7[0].set_ylabel('L1 Distance from Analytical Optimum', fontsize=12, fontweight='bold')
subplot_axes_step7[0].set_title('Step 7: Portfolio Weight Convergence Analysis\n(Unconstrained Optimization)',
subplot_axes_step7[0].grid(True, alpha=0.3, linestyle='--')
subplot_axes_step7[0].legend(fontsize=10)

subplot_axes_step7[1].plot(simulation_counts, sharpe_errors_percentage, marker='s', linewidth=2.5, markersize=
subplot_axes_step7[1].set_xscale('log')
subplot_axes_step7[1].set_yscale('log')
subplot_axes_step7[1].set_xlabel('Number of Monte Carlo Simulations', fontsize=12, fontweight='bold')
subplot_axes_step7[1].set_ylabel('Sharpe Ratio Error (%)', fontsize=12, fontweight='bold')
subplot_axes_step7[1].set_title('Step 7: Sharpe Ratio Convergence Analysis\n(Unconstrained Optimization)', fon
subplot_axes_step7[1].grid(True, alpha=0.3, linestyle='--')
subplot_axes_step7[1].legend(fontsize=10)

plt.tight_layout()
plt.show()

print("\n" + "="*80)
print("STEP 7: KEY FINDINGS")
print("="*80)
print(f"\nUnconstrained Portfolio Optimization Results:")
print(f"  - Number of actively invested assets: {sum(1 for w in optimal_weights_step7 if w > 0.05)}")
print(f"  - Optimal Sharpe Ratio (Analytical): {optimal_sharpe_step7:.6f}")
```

turnitin Page 28 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
print(f"  - Optimal Annual Return: {optimal_return_step7*100:.2f}%")
print(f"  - Optimal Annual Volatility: {optimal_volatility_step7*100:.2f}%")

convergence_threshold_1pct = None
for result_entry in convergence_results_step7:
    if result_entry['Sharpe_Error_Pct'] <= 1.0:
        convergence_threshold_1pct = result_entry['Simulations']
        break

if convergence_threshold_1pct:
    print(f"\nSimulations Required for 1% Sharpe Ratio Error: {convergence_threshold_1pct:,}")
else:
    print(f"\nSimulations Required for 1% Sharpe Ratio Error: Not achieved in tested range")

print("\n" + "="*80)
print("STEP 8: BOX-CONSTRAINED OPTIMIZATION VS SIMULATION")
print("="*80)

maximum_weight_limit = 0.30
minimum_weight_limit = 0.0

constraints_step8 = [
    {'type': 'eq', 'fun': lambda weights: np.sum(weights) - 1.0},
]
bounds_step8 = tuple((minimum_weight_limit, maximum_weight_limit) for _ in range(num_assets))

optimization_solution_step8 = minimize(
    negative_sharpe_objective,
    initial_weight_guess,
    args=(annual_expected_returns, covariance_matrix_annual),
    method='SLSQP',
    bounds=bounds_step8,
    constraints=constraints_step8,
    options={'ftol': 1e-12, 'maxiter': 1000}
)

optimal_weights_step8 = optimization_solution_step8.x
optimal_return_step8, optimal_volatility_step8, optimal_sharpe_step8 = portfolio_performance_metrics(
    optimal_weights_step8, annual_expected_returns, covariance_matrix_annual
```

turnitin Page 28 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
)

print("\nBox Constraint Applied: Maximum Weight per Asset = 30%")
print("\nOptimal Portfolio Weights (Optimization with Box Constraints):")
for ticker_idx, ticker_name in enumerate(asset_universe):
    print(f"  {ticker_name}: {optimal_weights_step8[ticker_idx]:.6f} ({optimal_weights_step8[ticker_idx]*100:.
print(f"\nOptimal Portfolio Annual Return: {optimal_return_step8:.6f} ({optimal_return_step8*100:.2f}%)")
print(f"Optimal Portfolio Annual Volatility: {optimal_volatility_step8:.6f} ({optimal_volatility_step8*100:.2f
print(f"Optimal Sharpe Ratio: {optimal_sharpe_step8:.6f}")

simulation_sample_sizes_step8 = [500000]
convergence_results_step8 = []

np.random.seed(42)

for sample_size in simulation_sample_sizes_step8:
    valid_portfolio_count = 0
    valid_weights_list = []
    valid_sharpe_list = []

    attempts = 0
    max_attempts = 50

    while valid_portfolio_count < sample_size and attempts < max_attempts:
        candidate_weights_matrix = np.random.dirichlet(np.ones(num_assets), sample_size * 3)

        for candidate_weight_vector in candidate_weights_matrix:
            if np.all(candidate_weight_vector <= maximum_weight_limit) and np.all(candidate_weight_vector >= m
                valid_weights_list.append(candidate_weight_vector)
                cand_return, cand_volatility, cand_sharpe = portfolio_performance_metrics(
                    candidate_weight_vector, annual_expected_returns, covariance_matrix_annual
                )
                valid_sharpe_list.append(cand_sharpe)
                valid_portfolio_count += 1

                if valid_portfolio_count >= sample_size:
                    break

        attempts += 1
```

```
    best_sim_index_s8 = np.argmax(valid_sharpe_list)
    best_simulated_weights_s8 = np.array(valid_weights_list[best_sim_index_s8])
    best_sim_return_s8, best_sim_volatility_s8, best_sim_sharpe_s8 = portfolio_performance_metrics(
        best_simulated_weights_s8, annual_expected_returns, covariance_matrix_annual
    )

    weight_l1_distance_s8 = np.sum(np.abs(best_simulated_weights_s8 - optimal_weights_step8))
    sharpe_absolute_difference_s8 = np.abs(best_sim_sharpe_s8 - optimal_sharpe_step8)
    sharpe_percentage_deviation_s8 = (sharpe_absolute_difference_s8 / optimal_sharpe_step8) * 100 if optimal_s

    convergence_results_step8.append({
        'Simulations': sample_size,
        'Optimal_Weights': best_simulated_weights_s8,
        'Portfolio_Return': best_sim_return_s8,
        'Portfolio_Volatility': best_sim_volatility_s8,
        'Sharpe_Ratio': best_sim_sharpe_s8,
        'L1_Distance': weight_l1_distance_s8,
        'Sharpe_Difference': sharpe_absolute_difference_s8,
        'Sharpe_Error_Pct': sharpe_percentage_deviation_s8
    })

    print(f"\nValid Simulations Generated: {sample_size}")
    print(f"  Best Simulated Weights: {[f'{w:.6f}' for w in best_simulated_weights_s8]}")
    print(f"  Best Simulated Sharpe Ratio: {best_sim_sharpe_s8:.6f}")
    print(f"  L1 Distance from Optimal: {weight_l1_distance_s8:.8f}")
    print(f"  Sharpe Ratio Difference: {sharpe_absolute_difference_s8:.8f} ({sharpe_percentage_deviation_s8:.4

convergence_table_step8 = pd.DataFrame([
    {
        'Sample_Size': r['Simulations'],
        'L1_Weight_Distance': r['L1_Distance'],
        'Sharpe_Error_Percentage': r['Sharpe_Error_Pct']
    }
    for r in convergence_results_step8
])

print("\n" + "-"*80)
print("STEP 8: CONVERGENCE ANALYSIS (WITH BOX CONSTRAINTS)")
```

turnitin Page 31 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
print('-'*80)
print(convergence_table_step8.to_string(index=False))

figure_step8, subplot_axes_step8 = plt.subplots(1, 2, figsize=(14, 5))

simulation_counts_s8 = [r['Simulations'] for r in convergence_results_step8]
weight_distances_s8 = [r['L1_Distance'] for r in convergence_results_step8]
sharpe_errors_percentage_s8 = [r['Sharpe_Error_Pct'] for r in convergence_results_step8]

subplot_axes_step8[0].scatter(simulation_counts_s8, weight_distances_s8, s=350, marker='D', color='darkgreen',
subplot_axes_step8[0].set_xscale('log')
subplot_axes_step8[0].set_yscale('log')
subplot_axes_step8[0].set_xlabel('Valid Constrained Portfolios Evaluated', fontsize=12, fontweight='bold')
subplot_axes_step8[0].set_ylabel('L1 Distance from Analytical Optimum', fontsize=12, fontweight='bold')
subplot_axes_step8[0].set_title('Step 8: Weight Convergence Validation\n(Box-Constrained: Max 30% per Asset)',
subplot_axes_step8[0].grid(True, alpha=0.3, linestyle='--')
subplot_axes_step8[0].legend(fontsize=10)

subplot_axes_step8[1].scatter(simulation_counts_s8, sharpe_errors_percentage_s8, s=350, marker='D', color='dar
subplot_axes_step8[1].set_xscale('log')
subplot_axes_step8[1].set_yscale('log')
subplot_axes_step8[1].set_xlabel('Valid Constrained Portfolios Evaluated', fontsize=12, fontweight='bold')
subplot_axes_step8[1].set_ylabel('Sharpe Ratio Error (%)', fontsize=12, fontweight='bold')
subplot_axes_step8[1].set_title('Step 8: Sharpe Error Validation\n(Box-Constrained: Max 30% per Asset)', fonts
subplot_axes_step8[1].grid(True, alpha=0.3, linestyle='--')
subplot_axes_step8[1].legend(fontsize=10)

plt.tight_layout()
plt.show()

figure_comparison, subplot_axes_comparison = plt.subplots(1, 2, figsize=(16, 6))

subplot_axes_comparison[0].plot(simulation_counts, sharpe_errors_percentage, marker='o', linewidth=3.5, marker
subplot_axes_comparison[0].scatter([500000], sharpe_errors_percentage_s8, s=450, marker='s', label='Step 8 Box
subplot_axes_comparison[0].set_xscale('log')
subplot_axes_comparison[0].set_yscale('log')
subplot_axes_comparison[0].set_xlabel('Number of Simulations / Valid Portfolios', fontsize=13, fontweight='bol
subplot_axes_comparison[0].set_ylabel('Sharpe Ratio Error (%)', fontsize=13, fontweight='bold')
subplot_axes_comparison[0].set_title('Convergence Comparison: Unconstrained vs Box-Constrained\nMonte Carlo Va
```

turnitin Page 31 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 32 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
subplot_axes_comparison[0].legend(fontsize=11, loc='upper right', framealpha=0.97)
subplot_axes_comparison[0].grid(True, alpha=0.35, linestyle='--', linewidth=1)
subplot_axes_comparison[0].set_xlim(5000, 2000000)

x_position_assets = np.arange(len(asset_universe))
bar_width = 0.38

subplot_axes_comparison[1].bar(x_position_assets - bar_width/2, optimal_weights_step7, bar_width, label='Step
subplot_axes_comparison[1].bar(x_position_assets + bar_width/2, optimal_weights_step8, bar_width, label='Step

subplot_axes_comparison[1].axhline(y=0.30, color='black', linestyle='--', linewidth=2.8, label='Maximum Constr
subplot_axes_comparison[1].set_ylabel('Portfolio Weight Allocation', fontsize=13, fontweight='bold')
subplot_axes_comparison[1].set_xlabel('Asset', fontsize=13, fontweight='bold')
subplot_axes_comparison[1].set_title('Optimal Allocation Comparison\nUnconstrained vs Box-Constrained', fontsi
subplot_axes_comparison[1].set_xticks(x_position_assets)
subplot_axes_comparison[1].set_xticklabels(asset_universe, fontsize=12, fontweight='bold')
subplot_axes_comparison[1].legend(fontsize=11, loc='upper right', framealpha=0.97)
subplot_axes_comparison[1].grid(True, alpha=0.3, axis='y', linestyle='--', linewidth=1)
subplot_axes_comparison[1].set_ylim(0, 0.42)

plt.tight_layout()
plt.show()

print("\n" + "="*80)
print("STEP 8: KEY FINDINGS")
print("="*80)
print(f"\nBox-Constrained Portfolio Optimization Results:")
print(f"  - All weights constrained to [0%, 30%] range")
print(f"  - Optimal Sharpe Ratio (Analytical): {optimal_sharpe_step8:.6f}")
print(f"  - Optimal Annual Return: {optimal_return_step8*100:.2f}%")
print(f"  - Optimal Annual Volatility: {optimal_volatility_step8*100:.2f}%")
print(f"  - Return Change vs Unconstrained: {((optimal_return_step8/optimal_return_step7 - 1)*100):.2f}%")

convergence_threshold_1pct_s8 = None
for result_entry in convergence_results_step8:
    if result_entry['Sharpe_Error_Pct'] <= 1.0:
        convergence_threshold_1pct_s8 = result_entry['Simulations']
        break
```

turnitin Page 32 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 33 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
if convergence_threshold_1pct_s8:
    print(f"\nSimulations Required for 1% Sharpe Ratio Error: {convergence_threshold_1pct_s8:,}")
else:
    print(f"\nSimulations Required for 1% Sharpe Ratio Error: Not achieved in tested range")

print(":-) "*5)
```

turnitin Page 33 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 34 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369
turnitin Page 34 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 35 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
================================================================================
STEP 7: UNCONSTRAINED OPTIMIZATION VS SIMULATION
================================================================================

Asset Universe: ['TSLA', 'WMT', 'BAC', 'LLY', 'AAPL']

Training Period: July 1, 2023 - June 30, 2025

Data Points Available: 520 trading days

Annualized Expected Returns:
  TSLA: 2.118054 (211.81%)
  WMT: -0.808438 (-80.84%)
  BAC: 0.372459 (37.25%)
  LLY: 1.888566 (188.86%)
  AAPL: 0.330331 (33.03%)

Annualized Covariance Matrix:
          TSLA       WMT       BAC       LLY      AAPL
TSLA  6.489085  0.283343  0.719839  1.395570  1.091522
WMT   0.283343  0.501867  0.217307  0.125899  0.200824
BAC   0.719839  0.217307  0.939397  0.361387  0.359580
LLY   1.395570  0.125899  0.361387  3.932990  0.701018
AAPL  1.091522  0.200824  0.359580  0.701018  1.173242

================================================================================
STEP 7: OPTIMIZATION APPROACH
================================================================================

Optimal Portfolio Weights (Optimization):
  TSLA: 0.339745 (33.97%)
  WMT: 0.000000 (0.00%)
  BAC: 0.095501 (9.55%)
  LLY: 0.564753 (56.48%)
  AAPL: 0.000000 (0.00%)

Optimal Portfolio Annual Return: 1.821743 (182.17%)
Optimal Portfolio Annual Volatility: 1.622724 (162.27%)
Optimal Sharpe Ratio: 1.122645

Simulation Count: 10000
Best Simulated Weights: ['0.356669', '0.009471', '0.007640', '0.581342', '0.044877']
```

turnitin Page 35 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

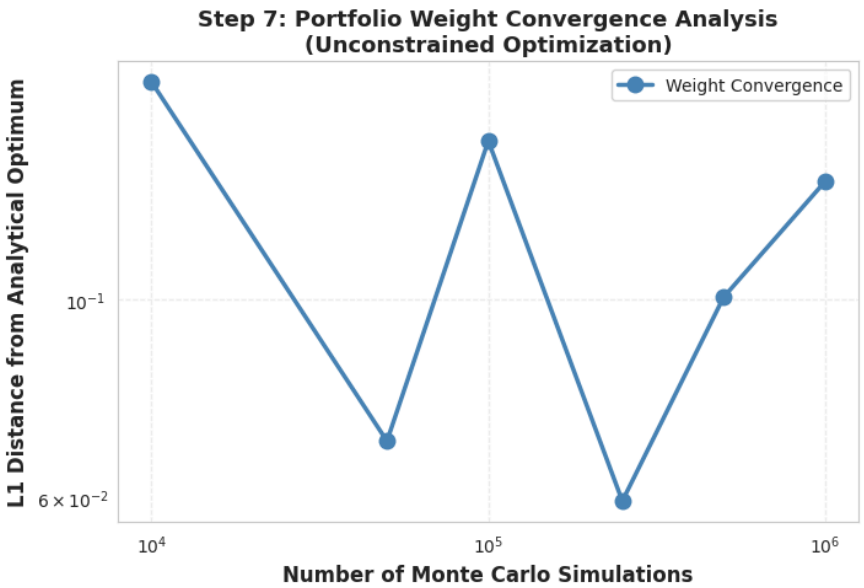
turnitin Page 35 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

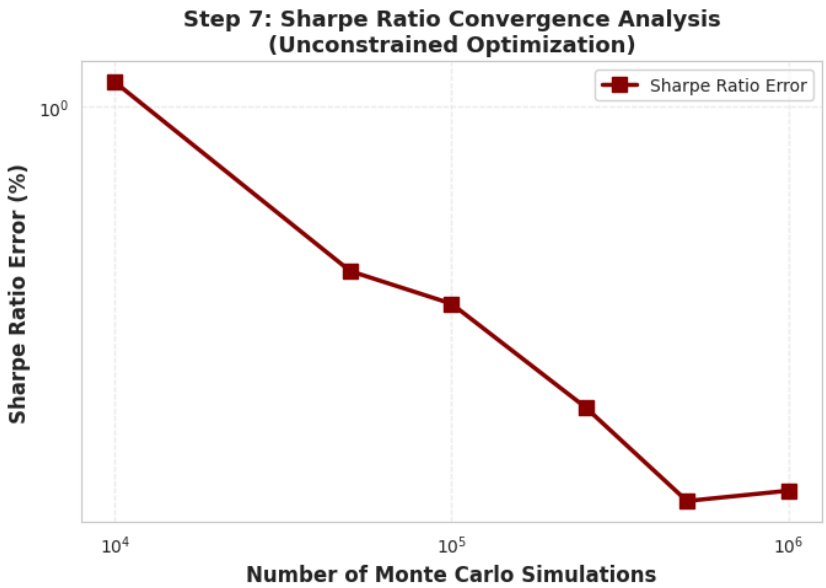
```
  Best Simulated Sharpe Ratio: 1.109953
  L1 Distance from Optimal: 0.17572212
  Sharpe Ratio Difference: 0.01269219 (1.1306%)

Simulation Count: 50000
  Best Simulated Weights: ['0.347064', '0.004888', '0.060808', '0.573275', '0.013966']
  Best Simulated Sharpe Ratio: 1.117746
  L1 Distance from Optimal: 0.06938688
  Sharpe Ratio Difference: 0.00489965 (0.4364%)

Simulation Count: 100000
  Best Simulated Weights: ['0.382537', '0.001844', '0.020070', '0.582389', '0.013160']
  Best Simulated Sharpe Ratio: 1.118489
  L1 Distance from Optimal: 0.15086200
  Sharpe Ratio Difference: 0.00415599 (0.3702%)

Simulation Count: 250000
  Best Simulated Weights: ['0.331486', '0.002302', '0.074050', '0.586614', '0.005548']
  Best Simulated Sharpe Ratio: 1.120178
  L1 Distance from Optimal: 0.05942033
  Sharpe Ratio Difference: 0.00246712 (0.2198%)

Simulation Count: 500000
  Best Simulated Weights: ['0.317306', '0.001465', '0.143771', '0.536831', '0.000627']
  Best Simulated Sharpe Ratio: 1.121106
  L1 Distance from Optimal: 0.10072535
  Sharpe Ratio Difference: 0.00153937 (0.1371%)

Simulation Count: 1000000
  Best Simulated Weights: ['0.350348', '0.000166', '0.027587', '0.619963', '0.001937']
  Best Simulated Sharpe Ratio: 1.121024
  L1 Distance from Optimal: 0.13582932
  Sharpe Ratio Difference: 0.00162143 (0.1444%)

--------------------------------------------------------------------------------
STEP 7: CONVERGENCE ANALYSIS
--------------------------------------------------------------------------------
 Sample_Size  L1_Weight_Distance  Sharpe_Error_Percentage
       10000            0.175722                 1.130561
       50000            0.069387                 0.436438
      100000            0.150862                 0.370196
      250000            0.059420                 0.219760
      500000            0.100725                 0.137120
```

turnitin Page 36 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

turnitin Page 37 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369

```
1000000          0.135829          0.144429
```






```
================================================================================
STEP 7: KEY FINDINGS
================================================================================

Unconstrained Portfolio Optimization Results:
  - Number of actively invested assets: 3
  - Optimal Sharpe Ratio (Analytical): 1.122645
  - Optimal Annual Return: 182.17%
  - Optimal Annual Volatility: 162.27%

Simulations Required for 1% Sharpe Ratio Error: 50,000
```

turnitin Page 37 of 37 - Integrity Submission Submission ID trn:oid:::8628:554317369